

THE MUSIC ENCODING INITIATIVE

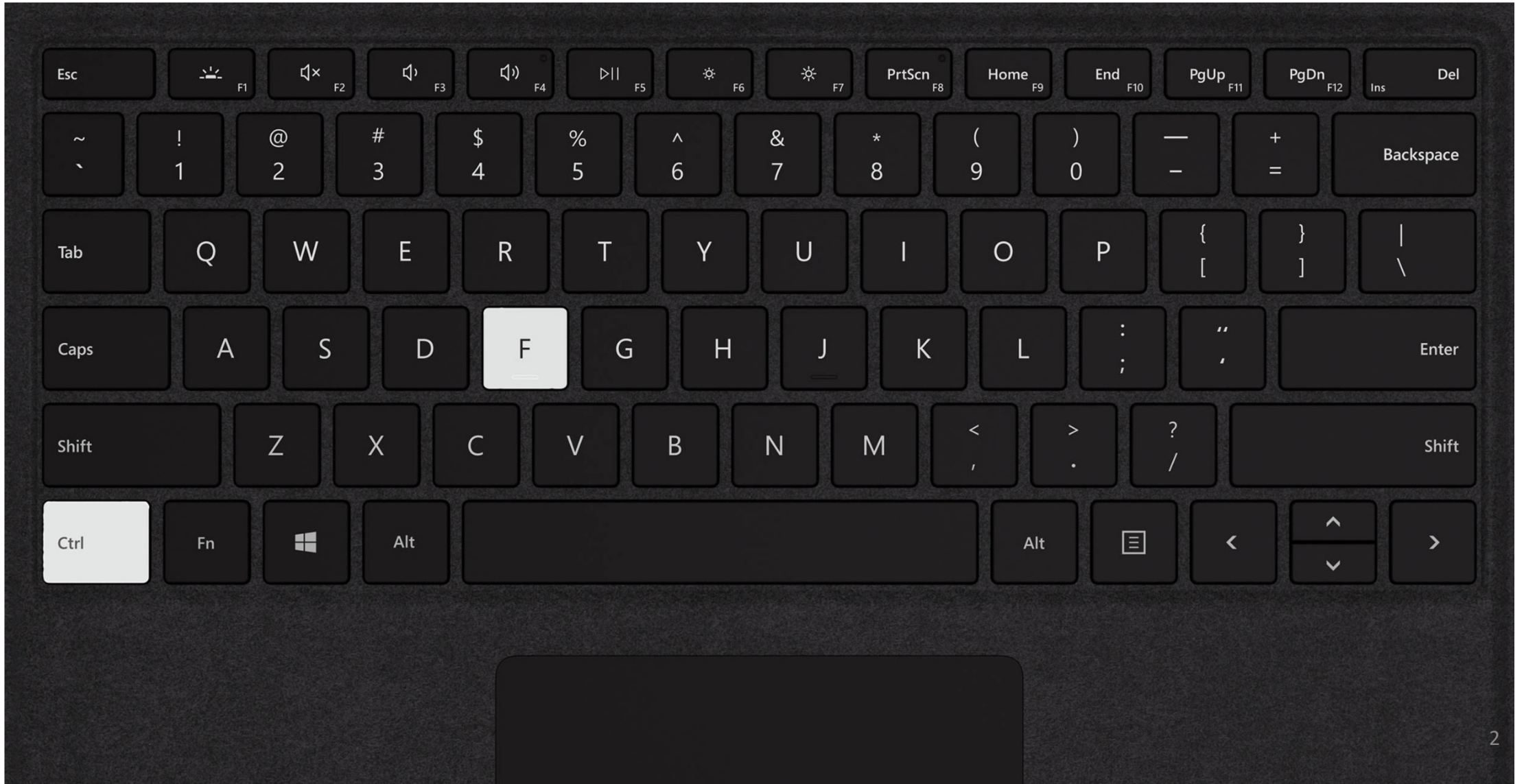
ENCODING OF EARLY MUSIC

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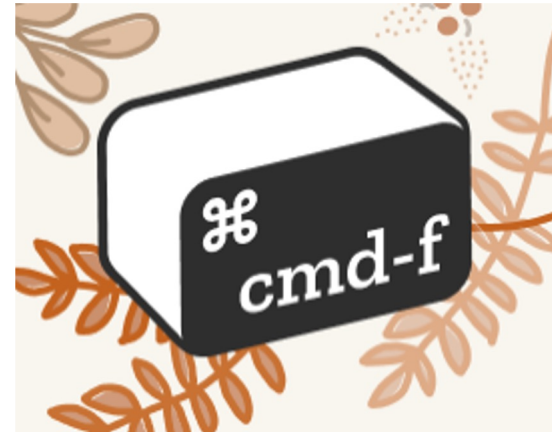
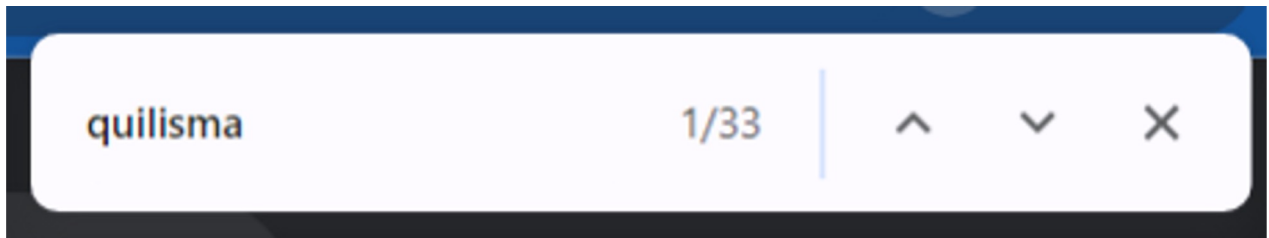
www.tinyurl.com/4jjtkpu7



ribi regnanti. p[ro]p[ri]um. ecc[lesi]e. m[ag]is. **G**la[ria] laus
 lebi. sancte. i[st]e. uenit. sancta. i[n] deu[m]. **G**la[ria]
 cum uero. i[n] carmine. digne. fere[n]t. **G**la[ria]
 laus. **R**egrediente dno
Mentem familie tue. **O**
 q[ui]s d[omi]ne intercedente beato. al.
 i[m]munere cōpunctionis ap[er]it. lar
 gitate pietatis exaudi. **P.** **Adm[is]**
 sam. **offm**
OMN
NRO
FALAS
 a. v. l. i. n. q. r. e. x. m. a. q. u. e. a. d. e. f. e. r. e. t. o.
 n. e. m. e. a. n. a. s. p. e. c. t. e. l. i. b. e. r. a. m. e. d. e. o. r. e. l. e. o. n. g.
 i. a. c. o. r. d. i. b. u. s. u. n. c. o. r. d. i. u. m. h. u. m. i. l. i. t. a. t. e. m. e.
 a. m. **p** d. e. u. s. d. e. u. s. i. s. t. a. s. e. u. a. c. e. **O**
Qu[od] p[er] sēptimē d[omi]n[u]s qui
 hūmāno generi ad imitā
 dū hūmilitatis exemplū
 saluātorem n[ost]r[u]m carnem sūmē

i[n] crucem subire fecisti. concede
 p[ro]p[ri]i ut i[n] patientie ip[s]i habere
 documēta. i[n] resurrectionis ei[us] con
 sortia meream[ur]. **P.** **ad philipenses.**
Rs. **H**oc sentite
Fim uob[is]. q[uo]d i[n] x[p]o ihu. q[ui]
 cū in forma dei ēēt nō ra
 pinā arbitrat[us] est. q[uo]d se equalē
 deo. sed semet ip[s]ū exinaniuit
 formā serui accipien[tes]. in similitu
 dinē hominū factus. i[n] habitu
 inuent[us] ut homo. humiliavit
 semet ip[s]ū factus obediens usq[ue]
 ad mortē. mortē autē crucis. q[ui]
 q[uo]d i[n] d[omi]n[o] exaltauit illū. i[n] donauit
 illi nom[en] q[uo]d ē sup[er] om[n]e nom[en]. ut in
 nomine ihu om[n]e genu flectat[ur].
 celestium terestrium i[n] inferno[rum].
 q[uo]d om[n]is lingua confiteatur. quia
 d[omi]n[u]s ihu x[p]c in gl[ori]a ē dei patris.

Constitui me in me a
 m[u]ltitudine in a
 dedisti me
 er cum gloria
 assumpti me
 uam bonis affat[us] de
 u
 re. d[omi]n[u]s
 r[e]le. m[ag]is autem
 gine mo
 re. sunt po
 de s[an]c[t]e pene



ribi regnanti. p[ro]p[ri]um. ecc[lesi]e. m[ag]is. **G**la[ria] laus
 lebi. sancte. i[st]e. uenit. sancta. i[n] deu[m]. **G**la[ria]
 cum uero. i[n] carmine. digne. fere[n]t. **G**la[ria]
 laus. **R**egrediente dno
Mentem familie tue. **O**
 q[ui]s d[omi]ne. intercedente beato. al.
 i[m]munere cōpunctionis. ap[er]it. lar
 gitate pietatis. exaudi. **P**. ad m[ag]is.
 lam. **O**ffm
OMNINO FALS
 a. viliq[ue]. reg[is]. a. me. ad. def[er]re
 nem. me. an. as[er]ue. libera. me. de. ore. leon[is]
 i[n] a. corib[us]. uice. p[ri]ncipum. humilitate. m[ag]is
 am. **P**. d[omi]ne. deus. d[omi]ne. **G**la[ria] s[an]c[t]i. u[er]bi. **O**
Qu[od] p[er] s[an]c[t]i. p[er] s[an]c[t]i. d[omi]ne. qui
 humano. generi. ad. imita
 dū. humilitatis. exempli
 saluatore[m]. n[ost]r[um]. carnem. sume

i[n] crucem subne fecisti. concede
 p[ro]p[ri]a ut i[n] pacientie ip[s]i habere
 docum[en]ta. i[n] resurrectionis ei[us] con
 sortia meream[ur]. **P**. ad philipenses.
Rs. **H**oc sentite
Fim uob[is]. q[uo]d i[n] x[p]o ihu. **Q**
 cū in forma dei e[ss]et nō ra
 pinā arbitrat[us] est. q[uo]d se equalē
 deo. sed semet ip[s]ū exinaniuit
 formā serui accipien[tes]. in similitu
 dinē hominū factus. i[n] habitu
 inuent[us] ut homo. humiliavit
 semet ip[s]ū factus obediens usq[ue]
 ad mortē. mortē autē crucis. **H**
 q[uo]d i[n] d[omi]no exaltauit illū. i[n] donauit
 illi nom[en] q[uo]d ē sup[er] om[n]e nom[en]. ut in
 nomine ihu om[n]e genu flectat[ur].
 celestium. terrestrium. i[n] inferno[rum].
 Q[uo]m[n]is lingua confiteatur. quia
 d[omi]n[us] ihu x[p]c in gl[ori]a ē dei patris.

Constitui me. in me. a
 in uoluntate. m[ag]is. de. d[omi]no. me
 er. cum. gloria. **C**onstitui me
 uam. bonis. affat[us]. de. u[er]bi
 re. d[omi]ne. co. r[ati]o. me. autem
 que. mo. re. sunt. po. de. s[an]c[t]i. p[er] ne

quillisma
 1/33
 ^
 v
 x



Optical Music Recognition (OMR) applied to a music manuscript would generate a file that contains the encoding of all the music information in the book. By means of this file, one can search for a neume shape in the blink of an eye.

OMR is the transformation of images of music notation into digital representations (i.e., machine-readable files) with limited direct human involvement.



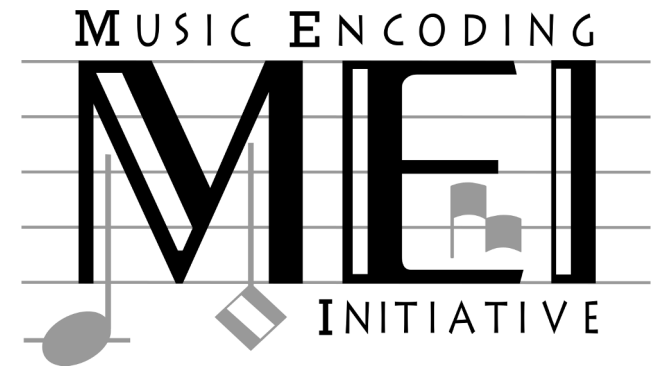
Why Would We Want to Encode Music?

Machine-readable files → Ask the computer:

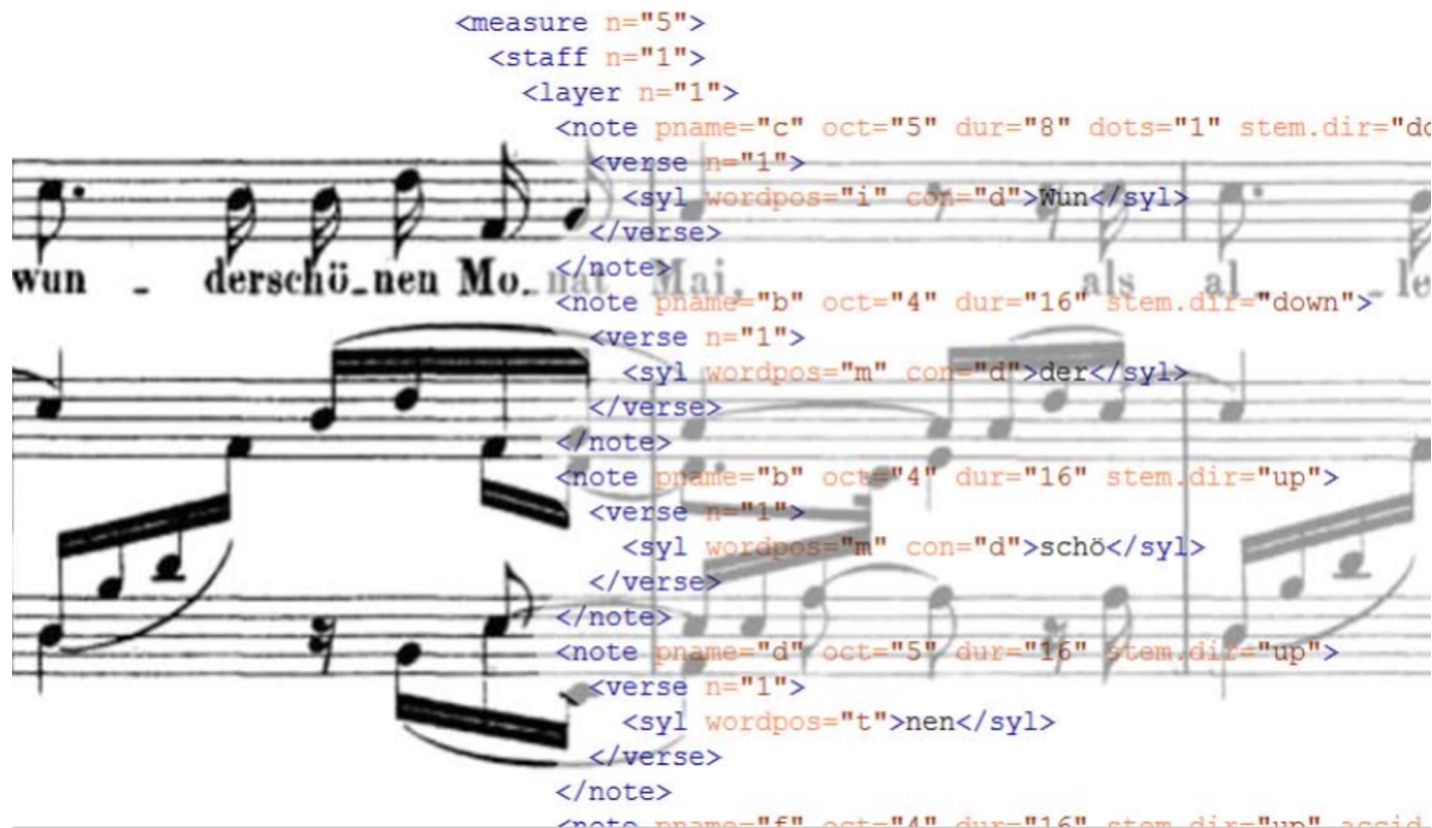
- Perform searches
 - Melodic search
 - Search for particular neume shapes
 - Look for melismatic passages
- Music analysis
 - Cross-comparison
 - Compute the mode of a chant based on features that the computer can extract, like *finalis*, *repercussio* (i.e., most repeated note), and *ambitus* (i.e., range)

Why Would We Want to Encode Music in MEI?

The Music Encoding Initiative (**MEI**), inspired by the Text Encoding Initiative (**TEI**), is a joint effort of an open and cosmopolitan community of technologists, musicologists, musicians, librarians, and music enthusiasts who discuss and approve rules aiming at fully capturing a wide variety of music documents into a machine-readable format

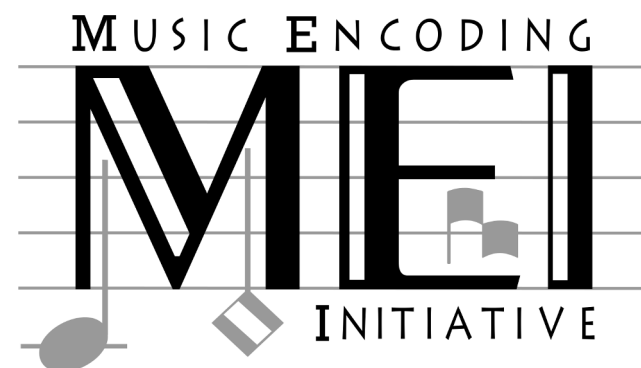


Why Would We Want to Encode Music in MEI?



The image shows a musical score with lyrics and MEI XML code overlaid. The lyrics are "wun - der schö - nen Mo - nat Mai, als al - le". The MEI XML code is as follows:

```
<measure n="5">
  <staff n="1">
    <layer n="1">
      <note pname="c" oct="5" dur="8" dots="1" stem.dir="down">
        <verse n="1">
          <syl wordpos="i" con="d">wun</syl>
        </verse>
      </note>
      <note pname="b" oct="4" dur="16" stem.dir="down">
        <verse n="1">
          <syl wordpos="m" con="d">der</syl>
        </verse>
      </note>
      <note pname="b" oct="4" dur="16" stem.dir="up">
        <verse n="1">
          <syl wordpos="m" con="d">schö</syl>
        </verse>
      </note>
      <note pname="d" oct="5" dur="16" stem.dir="up">
        <verse n="1">
          <syl wordpos="t">nen</syl>
        </verse>
      </note>
      <note pname="f" oct="4" dur="16" stem.dir="up" accid="f">
```



OMR recognizes the symbols on the page

MEI provides information on the *meaning* of the musical symbols

Why Would We Want to Encode Music in **MEI**?

MEI (or *Music Encoding Initiative*) is a machine-readable format that allows for the encoding of **early music** among other things:

- **Neume | Mensural | Tablature** (plus common Western music notation, CMN)
- **Rich metadata**
- **Relation to facsimiles & recordings**
- **Markup for Analysis and Harmonies**
- **Scholar editions**

Why Would We Want to Encode Music in **MEI**?

MEI (or *Music Encoding Initiative*) is a machine-readable format that allows for the encoding of **early music** among other things:

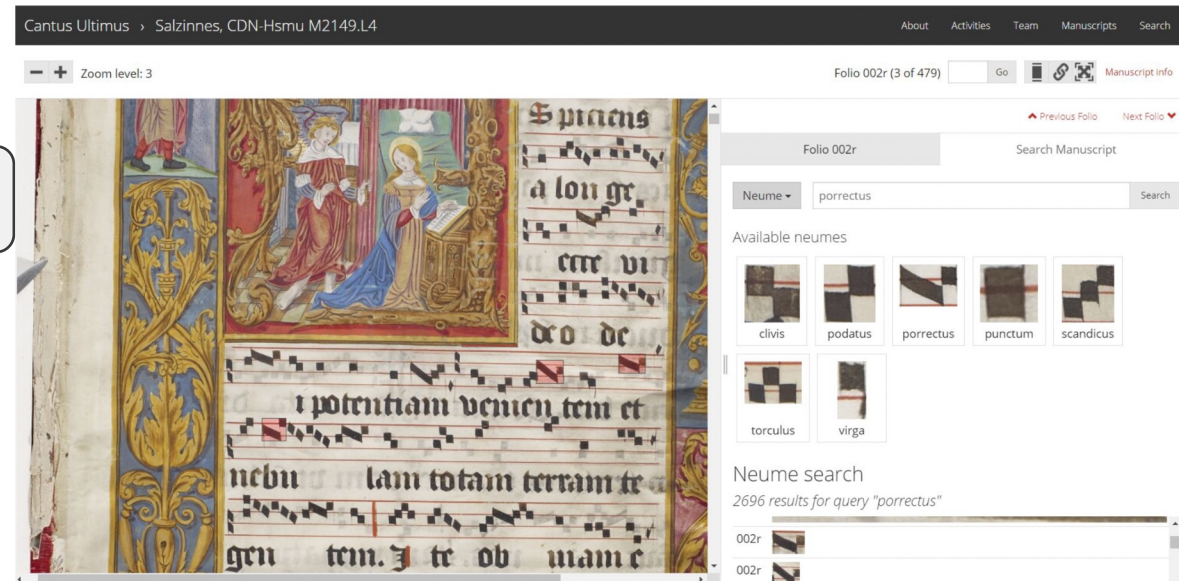
- **Neume | Mensural | Tablature** (plus common Western music notation, CMN)

- **Rich metadata**

- **Relation to facsimiles & recordings**

- **Markup for Analysis and Harmonies**

- **Scholar editions**

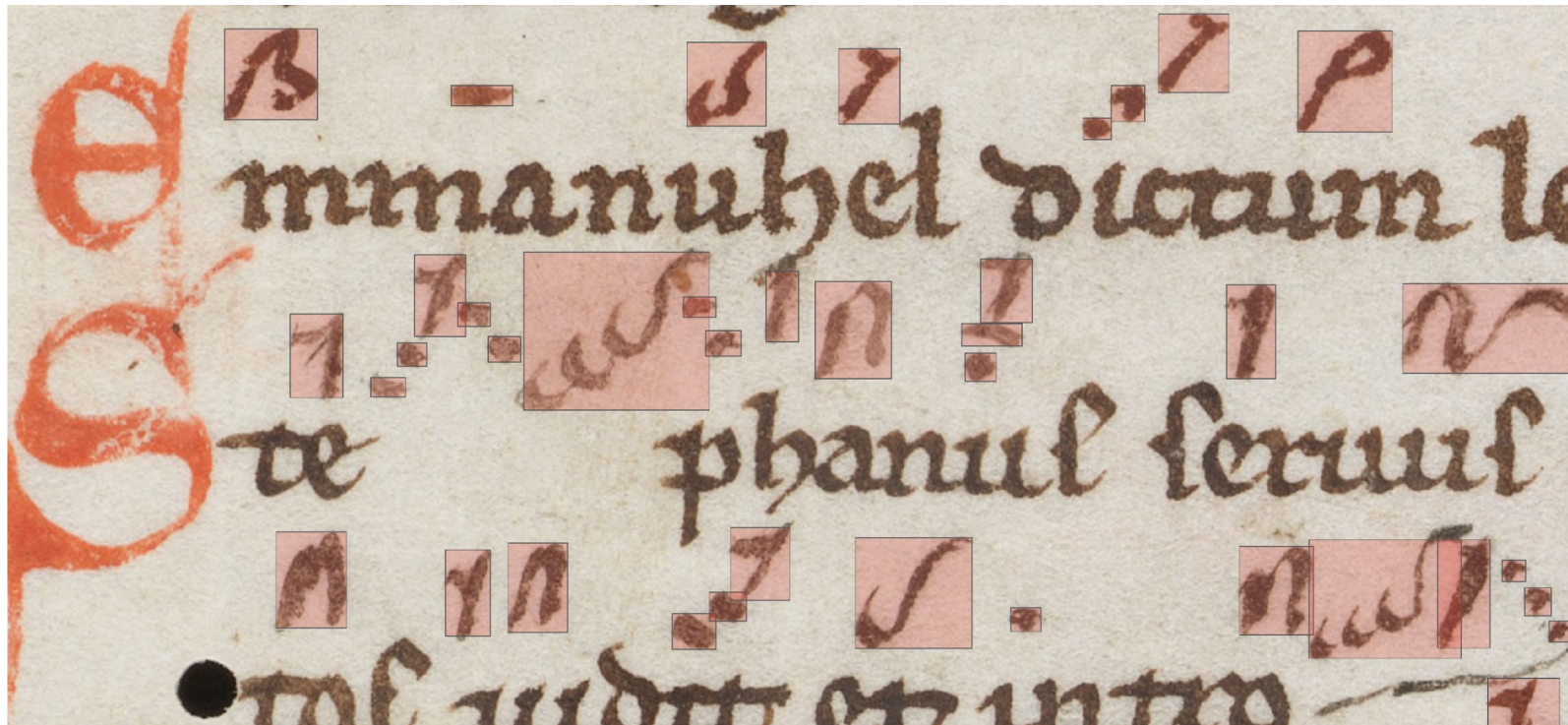


Two practical examples #1



```
<neume>  
  <nc curve= "a"/>  
  <nc tilt="ne" intm="d"/>  
</neume >
```

Two practical examples #1



Two practical examples #2

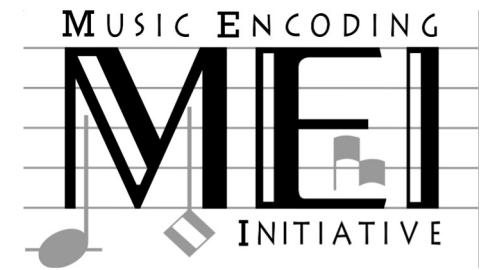


```
<neume>  
  <nc tilt="ne"/>  
  <nc tilt="se" relen="s" intm="d"/>  
</neume>
```

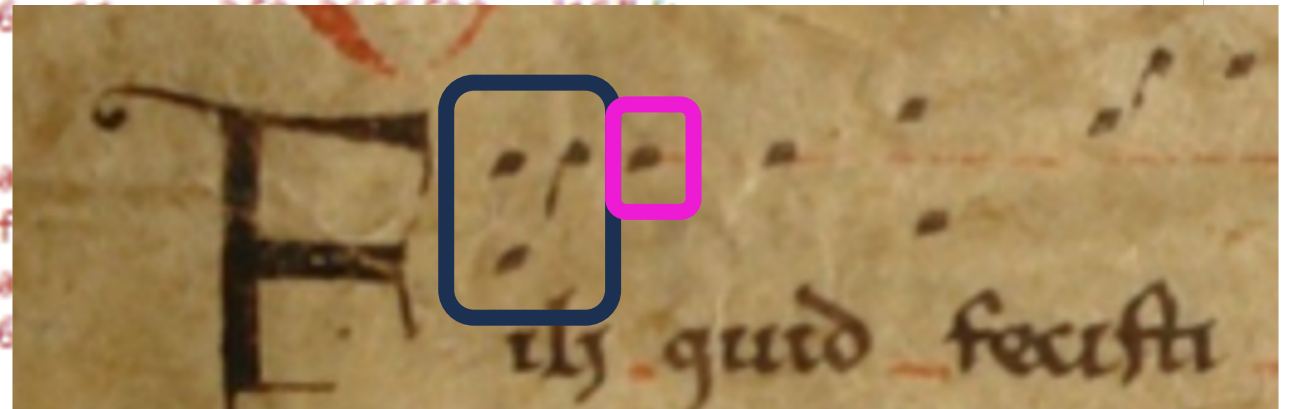


```
<neume>  
  <nc tilt="ne"/>  
  <nc tilt="se" relen="l" intm="d"/>  
</neume>
```


MEI is a hierarchical **system of tags** employed to describe in a machine-readable language the notation of a specific manuscript



```
<syllable xml:id="m-43558eee-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-43558f20-8a6a-11ee-a3f2-3645f29ecd15">Fi</syl>
  <neume xml:id="m-43558f52-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-43558f8e-8a6a-11ee-a3f2-3645f29ecd15"/>
    <nc loc="-2" xml:id="m-43558fc0-8a6a-11ee-a3f2-3645f29ecd15"/>
    <nc loc="0" tilt="ne" xml:id="m-43558ff2-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>
<syllable xml:id="m-43559024-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-4355904c-8a6a-11ee-a3f2-3645f29ecd15">li</syl>
  <neume xml:id="m-4355907e-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-435590b0-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>
<syllable xml:id="m-435590d8-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-4355910a-8a6a-11ee-a3f2-3645f29ecd15">
    <neume xml:id="m-4355913c-8a6a-11ee-a3f2-3645f29ecd15">
      <nc loc="0" xml:id="m-43559178-8a6a-11ee-a3f2-3645f29ecd15"/>
    </neume>
  </syl>
</syllable>
```



Short Introduction to MEI



Music Encoding Initiative (MEI)

- A community and a format
- **The format:** Encodes music notation in XML (*eXtensible Markup Language*)
 - Hierarchical encoding (tree structure)
 - Its core objects are **elements** (tags, e.g. `<note>`)
 - Opening tags & closing tags → when there is content in between
 - Self-closing tags → when there is no content to be included in between tags
 - Uses **attributes** to define the properties of elements
 - XML does not “do” anything, it is just information wrapped in tags
- **Goal:** Encode a wide variety of music documents

	<code><beam></code> Opening tag <code><note pname="f" oct="4" dur="8"/></code> Self-closing tag <code><note pname="c" oct="4" dur="8"/></code> <code></beam></code> Closing tag
--	--

XML is a hierarchical encoding system



```
<neume>  
  <nc tilt="e" />  
  <nc tilt="n" intm="u" />  
</neume>
```

<neume> is the parent element = Sign representing one or more musical pitches

<nc> is a child element = Sign representing a single pitched event, although the exact pitch may not be known

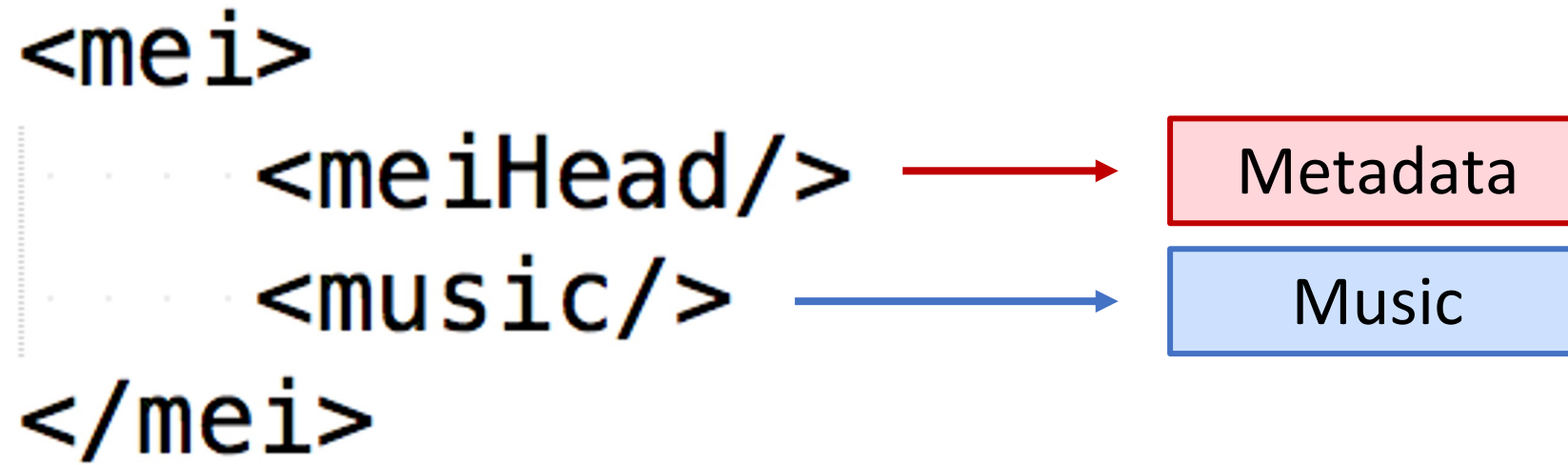
@tilt is an attribute of **<nc>** = Direction of the pen-stroke

@intm is an attribute of **<nc>** = Encodes the melodic interval from the previous neume component

Basic Structure of an MEI File



Basic Structure of an MEI File





Basic Structure of an MEI File

```
<mei>  
  . . . . . <meiHead/> —————→ Metadata  
  . . . . . <music/>  
</mei>
```



<meiHead> basic elements

```
<mei>
  <meiHead>
    <fileDesc>
      <titleStmt>
        <title></title>
      </titleStmt>
      <pubStmt/>
    </fileDesc>
  </meiHead>
  <music/>
</mei>
```

TUTORIALS
[https://music-encoding.org/
resources/tutorials.html](https://music-encoding.org/resources/tutorials.html)

XML Basics and Minimal MEI File Structure

Outermost basic structure of
an **MEI-conformant** document
(conforms to the **schema** of MEI)

Basic structure
of a “valid” MEI file

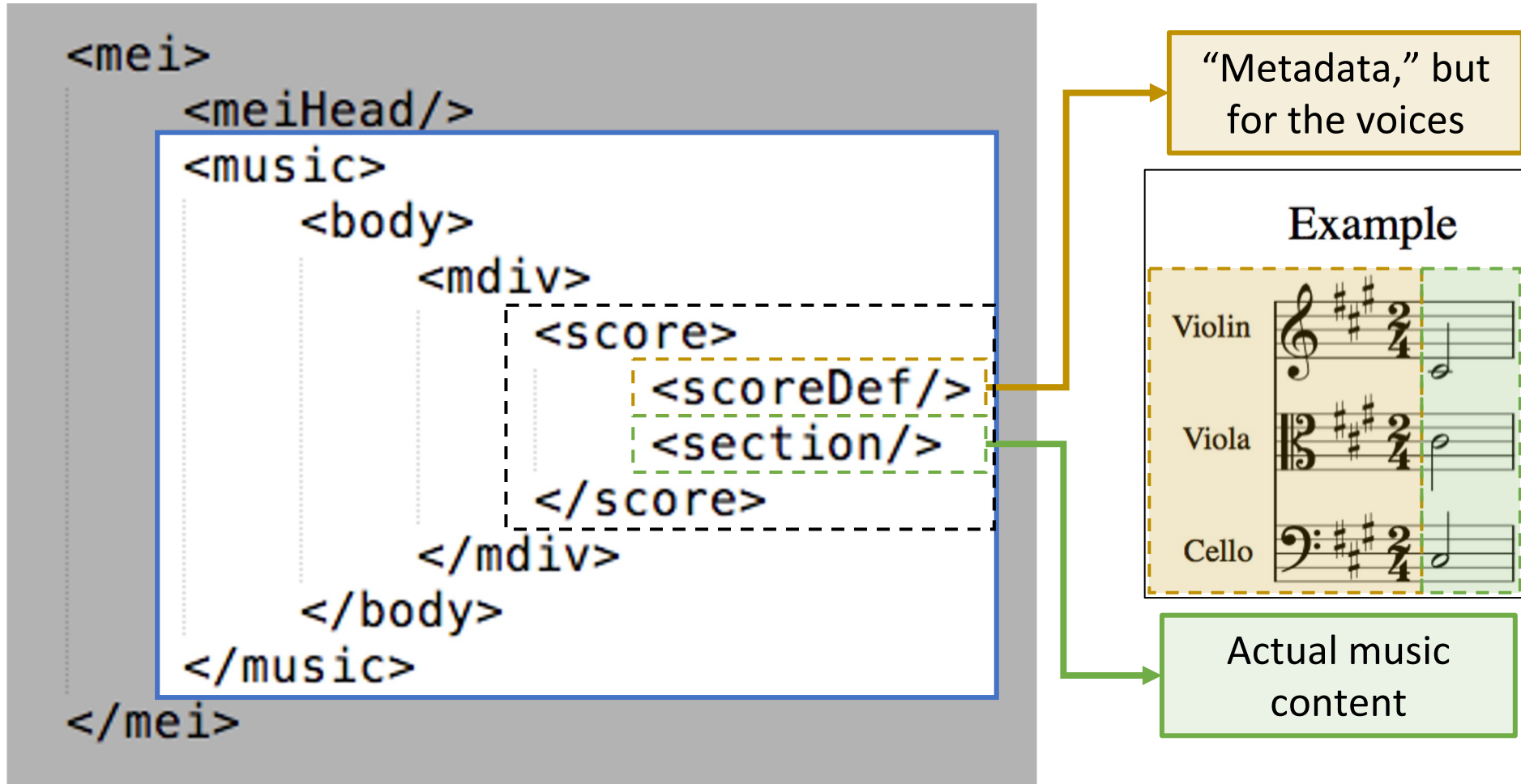


Basic Structure of an MEI File

```
<mei>  
  . . . <meiHead/>  
  . . . <music/> —————→ Music  
</mei>
```



<music> basic elements



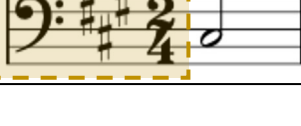


<music> basic elements

```
<mei>
  <meiHead/>
  <music>
    <body>
      <mdiv>
        <score>
          <scoreDef/>
          <section/>
        </score>
      </mdiv>
    </body>
  </music>
</mei>
```

“Metadata,” but
for the voices

Example

Violin	
Viola	
Cello	

<scoreDef> - General Information for Voices



```
<score>
  <scoreDef>
    <staffGrp>
      <staffDef n="1"/>
      <staffDef n="2"/>
      <staffDef n="3"/>
    </staffGrp>
  </scoreDef>
  <section/>
</score>
```

General information for **ALL** voices

- Meter
- Key

Specific information for **EACH** voice

- Clef
- Label
- **Number of staff-lines (@lines)**
- **@n**

required for validation

Example

The example shows three staves of musical notation. The top staff is labeled 'Violin' and uses a treble clef. The middle staff is labeled 'Viola' and uses an alto clef. The bottom staff is labeled 'Cello' and uses a bass clef. All three staves are in the key of D major (two sharps) and 2/4 time. The notation is enclosed in a dashed box, and the labels are in ovals.

<https://music-encoding.org/guidelines/v5/content/shared.html>



<music> basic elements

```
<mei>
  <meiHead/>
  <music>
    <body>
      <mdiv>
        <score>
          <scoreDef/>
          <section/>
        </score>
      </mdiv>
    </body>
  </music>
</mei>
```

Example

Violin

Viola

Cello

Actual music content



<section> - Actual Music

```
<score>
  <scoreDef/>
  <section>
    <measure n="1">
      <staff n="1">
        <layer>
          <!-- MUSIC WITHIN THE STAFF -->
        </layer>
      </staff>
      <staff n="2"/>
      <staff n="3"/>
    </measure>
    ...
    <measure n="20"/>
  </section>
</score>
```

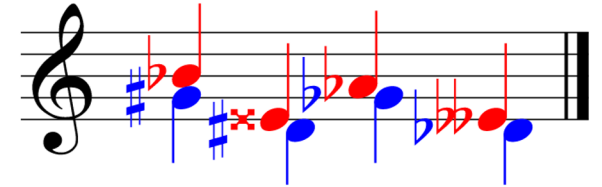
<https://music-encoding.org/guidelines/v5/content/shared.html>



<section> - Actual Music

```
<score>
  <scoreDef/>
  <section>
    <measure n="1">
      <staff n="1">
        <layer>
          <!-- MUSIC WITHIN THE STAFF -->
        </layer>
      </staff>
      <staff n="2"/>
      <staff n="3"/>
    </measure>
    ...
    <measure n="20"/>
  </section>
</score>
```

<layer n="1">



<layer n="2">

<https://music-encoding.org/guidelines/v5/content/shared.html>



Early Music has No Measures!

```
<score>
  <scoreDef/>
  <section>
    <measure n="1">
      <staff n="1">
        <layer>
          <!-- MUSIC WITHIN THE STAFF -->
        </layer>
      </staff>
      <staff n="2"/>
      <staff n="3"/>
    </measure>
    ...
    <measure n="20"/>
  </section>
</score>
```



Monophonic Music has a Single Staff

```
<score>
  <scoreDef/>
  <section>
    <staff n="1">
      <layer>
        <!-- MUSIC WITHIN THE STAFF -->
      </layer>
    </staff>
    <staff n="2"/>
    <staff n="3"/>
  </section>
</score>
```

Common Structure for Monophonic Music



```
<mei>
  <meiHead>
    <fileDesc>
      <titleStmt>
        <title></title>
      </titleStmt>
      <pubStmt/>
    </fileDesc>
  </meiHead>
  <music>
    <body>
      <mdiv>
        <score>
          <scoreDef/>
          <section/>
        </score>
      </mdiv>
    </body>
  </music>
</mei>
```

"Metadata,"
but for the voices

Example

Violin

Viola

Cello

```
<scoreDef>
  <staffGrp>
    <staffDef n="1"/>
    <staffDef n="2"/>
    <staffDef n="3"/>
  </staffGrp>
</scoreDef>
```

Actual music
content

```
<section>
  <staff n="1">
    <layer>
      <!-- MUSIC WITHIN THE STAFF -->
    </layer>
  </staff>
  <staff n="2"/>
  <staff n="3"/>
</section>
```

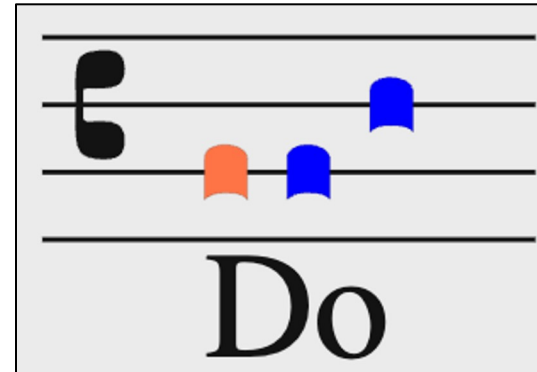

MEI for Neume Notation

Most neume notation is used to set music to an existing text. Therefore, the “syllable” is the fundamental unit of structure, with the neumes themselves serving as a means of “sonifying” the text.

Therefore, the syllable element provides high-level organization in this repertoire.

Elements

- `<syllable>`
 - **One** `<syl>` child
 - **One or more** `<neume>` children
 - neume component `<nc>` children, these are the individual notes that form part of the neume



```
<syllable>  
  <syl>Do</syl>  
  <neume>  
    <nc loc="2" />  
  </neume>  
  <neume>  
    <nc loc="2" />  
    <nc loc="4" />  
  </neume>  
</syllable>
```

Elements

- <syllable>
 - **One** <syl> child
 - **One or more** <neume> children
 - neume component <nc> children, these are the individual notes that form part of the neume

Attributes of neume components:

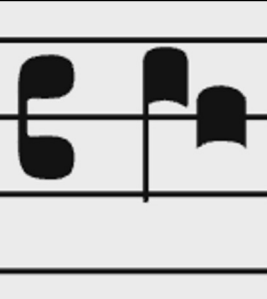
- **Pitch**
 - ❖ @pname = c | d | e | f | g | a | b
 - ❖ @oct = ... 2 | 3 | 4 ...
 - ❖ @loc: position in the staff (integer, 0 is the first line) = ... -2 | -1 | 0 | 1 | 2 ...
 - ❖ @intm: melodic interval (integer: negative, zero, positive) = ... -2 | -1 | 0 | 1 | 2 ...
- **Shape**
 - ❖ @tilt: direction of penstroke = n | s | se | ...
 - ❖ @curve = c | a
 - ❖ ...

Elements

- <syllable>
 - **One** <syl> child
 - **One or more** <neume> children
 - neume component <nc> children, these are the individual notes that form part of the neume

Attributes of neume components:

- **Pitch**
 - ❖ @pname = c | d | e | f | g | a | b
 - ❖ @oct = ... 2 | 3 | 4 ...
 - ❖ @loc: position in the staff (integer, 0 is the first line) = ... -2 | -1 | 0 | 1 | 2 ...
 - ❖ @intm: melodic interval (integer: negative, zero, positive) = ... -2 | -1 | 0 | 1 | 2 ...
- **Shape**
 - ❖ @tilt: direction of penstroke = n | s | se | ...

	<pre><neume> <nc pname="d" oct="3" tilt="n" /> <nc pname="c" oct="3" /> </neume></pre>
---	--

Elements

- <syllable>
 - **One** <syl> child
 - **One or more** <neume> children
 - neume component <nc> children, these are the individual notes that form part of the neume

Attributes of neume components:

○ Pitch

- ❖ @pname = c | d | e | f | g | a | b
- ❖ @oct = ... 2 | 3 | 4 ...
- ❖ @loc: position in the staff (integer, 0 is the first line) = ... -2 | -1 | 0 | 1 | 2 ...
- ❖ @intm: melodic interval (integer: negative, zero, positive) = ... -2 | -1 | 0 | 1 | 2 ...

○ Shape

- ❖ @tilt: direction of penstroke = n | s | se | ...
- ❖ @curve = c | a
- ❖ ...



```
<neume>  
  <nc loc="4" />  
  <nc intm="1" tilt="s" />  
</neume>
```

Elements

- `<syllable>`
 - **One** `<syl>` child
 - **One or more** `<neume>` children
 - neume component `<nc>` children, these are the individual notes that form part of the neume

Attributes of neume components:

- **Pitch**
 - ❖ `@pname` = c | d | e | f | g | a | b
 - ❖ `@oct` = ... 2 | 3 | 4 ...
 - ❖ `@loc`: position in the staff (integer, 0 is the first line) = ... -2 | -1 | 0 | 1 | 2 ...
 - ❖ `@intm`: melodic interval (integer: negative, zero, positive) = ... -2 | -1 | 0 | 1 | 2 ...
- **Shape**
 - ❖ `@tilt`: direction of penstroke = n | s | se | ...
 - ❖ `@curve` = c | a
 - ❖ ...



```
<neume>
  <nc tilt="e" />
  <nc tilt="n" intm="u" />
</neume>
```


Elements

- `<syllable>`
 - **One** `<syl>` child
 - **One or more** `<neume>` children
 - neume component `<nc>` children, these are the individual notes that form part of the neume



```
<neume>
  <nc tilt="ne" />
  <nc tilt="se" curve="c"
    intm="d" />
</neume>
```

Old Hispanic notation

Attributes of neume components:

- **Pitch**
 - ❖ `@pname` = c | d | e | f | g | a | b
 - ❖ `@oct` = ... 2 | 3 | 4 ...
 - ❖ `@loc`: position in the staff (integer, 0 is the first line) = ... -2 | -1 | 0 | 1 | 2 ...
 - ❖ `@intm`: melodic interval (integer: negative, zero, positive) = ... -2 | -1 | 0 | 1 | 2 ...
- **Shape**
 - ❖ `@tilt`: direction of penstroke = n | s | se | ...
 - ❖ `@curve` = c | a
 - ❖ ...

Elements

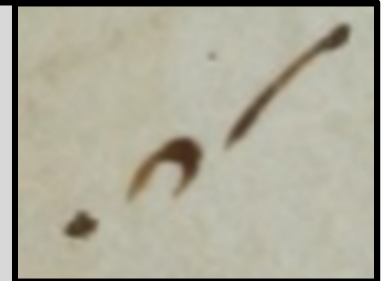
- **<syllable>**
 - **One** **<syl>** child
 - **One or more** **<neume>** children
 - neume component **<nc>** children, these are the individual notes that form part of the neume
 - **<nc>** can have children too:
 - **<episema>** ○ **<oriscus>**
 - **<hispanTick>** ○ **<quilisma>**
 - **<liquescent>** ○ **<strophicus>**

https://music-encoding.org/guidelines/v5/elements/nc.html#mayContain_class_tab

Elements

- `<syllable>`
 - **One** `<syl>` child
 - **One or more** `<neume>` children
 - neume component `<nc>` children, these are the individual notes that form part of the neume
 - `<nc>` can have children too:
 - `<episema>` ○ `<oriscus>`
 - `<hispanTick>` ○ `<quilisma>`
 - `<liquescent>` ○ `<strophicus>`

```
<neume>  
  <nc tilt="e"/>  
  <nc>  
    <oriscus/>  
  </nc>  
  <nc tilt="ne" intm="u"/>  
</neume>
```



St. Gall

https://music-encoding.org/guidelines/v5/elements/nc.html#mayContain_class_tab

Elements

- `<syllable>`
 - **One** `<syl>` child
 - **One or more** `<neume>` children
 - neume component `<nc>` children, these are the individual notes that form part of the neume
 - `<nc>` can have children too:
 - `<episema>` ○ `<oriscus>`
 - `<hispanTick>` ○ `<quilisma>`
 - `<liquescent>` ○ `<strophicus>`

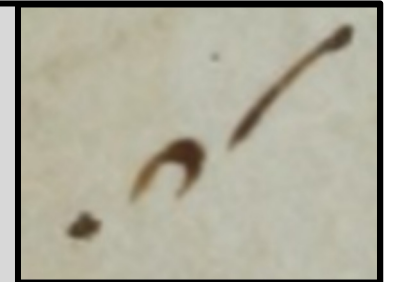
Attributes for `<episema>`:

- ❖ **@form** 'v' (vertical) or 'h' (horizontal)
- ❖ **@place** placement on `<nc>`

Attributes for `<liquescent>`:

- ❖ **@curve** direction of curve
- ❖ **@looped** boolean (when present, 'true')

```
<neume>
  <nc tilt="e"/>
  <nc>
    <oriscus/>
  </nc>
  <nc tilt="ne" intm="u"/>
</neume>
```



St. Gall

https://music-encoding.org/guidelines/v5/elements/nc.html#mayContain_class_tab

Attributes of neume components:

- ❖ **@pname** (pitch name) {[a-g] | unknown}
- ❖ **@oct** (octave number) {[0-9]}
- ❖ **@rellen** (length of pen stroke relative to previous <nc> in the neume) {s = shorter | l = longer}
- ❖ **@tilt** (direction of pen stroke) {n | ne | e | se | s | sw | w | nw}
- ❖ **@con** (connection to the previous <nc> in the neume) {g = gapped | l = looped | e = extended}
- ❖ **@ligated** {true | false}

Attributes of neume components in stafless notation:

- ❖ **@intm** (melodic interval relative to previous <nc> in the neume) {u = up | d = down | s = same}
- ❖ **@s-shape** (described by the direction of the initial penstroke) {n = north | s = south | ...}
- ❖ **@curve** (pen stroke) {c = clockwise | a = anti-clockwise}



```
<neume>  
  <nc tilt="ne"/>  
  <nc tilt="se" relen="s" intm="d"/>  
</neume>
```



```
<neume>  
  <nc tilt="ne"/>  
  <nc tilt="se" relen="l" intm="d"/>  
</neume>
```

Same melodic contour: Neutral – High – Low | Different neumatic connections

<neume>

<nc curve="a"/>

<nc tilt="ne" relen="l" intm="u"/>

<nc tilt="se" relen="s"
intm="d" curve="c"/>

</neume>



<neume>

<nc tilt="se"/>

<nc con="g" tilt="ne" relen="l"
intm="u"/>

<nc tilt="se" relen="l" intm="d"/>

</neume>



<neume>

<nc tilt="se"/>

<nc tilt="ne" relen="l" intm="u"/>

<nc tilt="se" intm="d"/>

</neume>



<neume>

<nc s-shaped="s"/>

<nc con="l" tilt="ne" intm="u"/>

<nc tilt="se" relen="s" intm="d"/>

</neume>



Let's Encode a Music Example

Using **mei-friend**: <https://mei-friend.mdw.ac.at/>



Editor panel
(MEI code)

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <?xml-model href="https://music-encoding.org/schema/5.0/mei-all.rng" type="application/xml" sc
3 <?xml-model href="https://music-encoding.org/schema/5.0/mei-all.rng" type="application/xml" sc
4 <mei xmlns="http://www.music-encoding.org/ns/mei" meiversion="5.0">
5   <meiHead xml:id="m-31bb19b6-abc0-4cbf-b0f6-169c67b5a315">
6     <fileDesc xml:id="m-01dadaef-3dc0-4521-a261-69333e36b1b3">
7       <titleStmt xml:id="m-bd9ae7c6-2c8d-48b8-ba82-4b94407d3c47">
8         <title xml:id="m-402a2186-79f9-4f25-a801-5acd1145497f">MEI Encoding Output (1.0.0)</title>
9       </titleStmt>
10      <pubStmt xml:id="m-6bf82541-43f0-47df-b1d4-0392281dba7b"/>
11    </fileDesc>
12    </meiHead>
13    <music xml:id="mg6szwz2">
14      <facsimile xml:id="m-4e57845f-8555-4122-b53f-ae59303a03a8">↔</facsimile>
34      <body xml:id="b02jngs1">
35        <mdiv xml:id="mopqc6s2">
36          <score xml:id="smc64">
37            <scoreDef xml:id="son6y6s2">
38              <staffGrp xml:id="syp81gz1">
39                <staffDef xml:id="skowclt1" clef.line="4" clef.shape="C" lines="5" n="1"
40              </staffGrp>
41            </scoreDef>
```

Rendering panel
(Verovio, rendering
engine for MEI
developed by
Laurent Pugin)



K < Page 1 of 1 > >I

Automatic

Update

Speed mode

MEI Encoding Output (1.0.0)



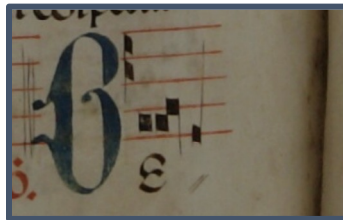
Alle lu ya **V** Sancte
 michael archagele defende nos in preli
 o ut nō pereamus in tremēdo iudicio. *Offi.*
Sicut angelus iuxta a
 ram templi habes thuribularia
 aurea in manu sua et data sunt ei
 incensamina et ascen
 dit fumus altarium in conspectu
 dei alle lu ya. *Co.* **G**

nedicite omnes angeli domini dominum hymnum dici
 te et super exalta te eum in secula.
Sci. Francisci. Inmtr. **A**lle lu ya
Gaudemus oēs. **S**icut frater cūctus pauper
& os iusti med. et modicus celum dives ingre
 ditur hymnus ecclestiasticus.
Offi. veritas mea. ed. fidelis seruus. **E**cce
Translato. S. Petri archiepi. Inmtr.
Statuit ei. et alia de cōt. vni. nris.
Festinas oim scōz. ut libro. Martini epi.
 sacerdos magnus martinus gemma sa
 cerdotum quem principem fecit domini ut sit no

*Sic luce sicut
 in festo scti
 Mathias.*

Let's start

- We will encode together a small example in MEI step by step
- We will use **mei-friend**, a web-based MEI editor
- Now, open the [skeleton MEI file](#) in mei-friend



Minimal MEI file structure

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-model href="https://music-encoding.org/schema/5.0/mei-Neumes.rng"
type="application/xml" schematypens="http://relaxng.org/ns/structure/1.0"?>
<?xml-model href="https://music-encoding.org/schema/5.0/mei-Neumes.rng"
type="application/xml" schematypens="http://purl.oclc.org/dsdl/schematron"?>
```

XML schema
definitions

```
<mei xmlns="http://www.music-encoding.org/ns/mei" meiversion="5.0">
```

Opening tag of root
element with
namespace declaration

```
<meiHead>
  <fileDesc>
    <titleStmt>
      <title>Basic MEI Mensural skeleton</title>
    </titleStmt>
    <pubStmt></pubStmt>
  </fileDesc>
</meiHead>
```

Header with file description and title
statement (all metadata goes here)

```
<music></music>
```

Music tag

```
</mei>
```

Closing tag of root element

The skeleton of the body of music for “neumes”



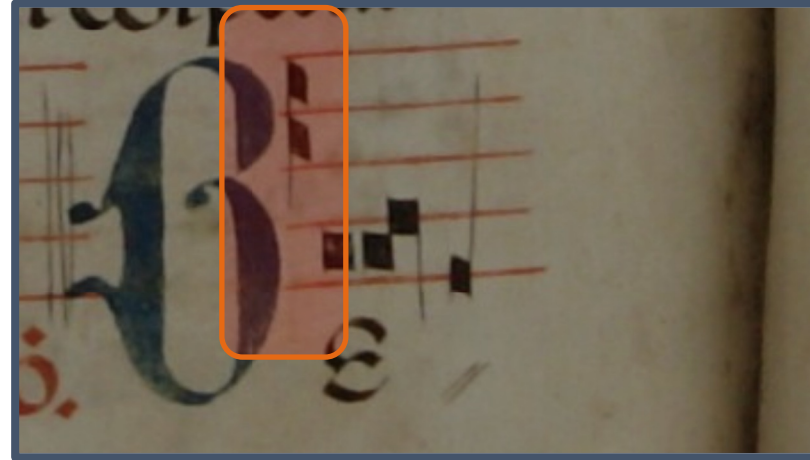
- The score definition `<scoreDef>` contains the definitions for the whole score, staff groups, and every single staff
- `<staffDef>` needs at least `@n` and `@lines`
- Use `@notationtype = neume` when dealing with neumatic notation
- `<section>` is used to separate music data segments
- `<staffDef>` and `<staff>` are linked by assigning the same number in `@n`
- A `<staff>` may contain several `<layer>` elements for multiple parts

```
13 <music>
14   <body>
15     <mdiv>
16       <score>
17         <scoreDef>
18           <staffGrp>
19             <staffDef n="1" lines="4" notationtype="neume"/>
20           </staffGrp>
21         </scoreDef>
22         <section>
23           <staff n="1">
24             <layer>
25               <!-- music content goes in here -->
26             </layer>
27           </staff>
28         </section>
29       </score>
30     </mdiv>
31   </body>
32 </music>
```

Add a clef in the staff definition



- Add clef attributes to `<staffDef>`:
 - Add `@clef.shape = C`
 - Add `@clef.line = 4`



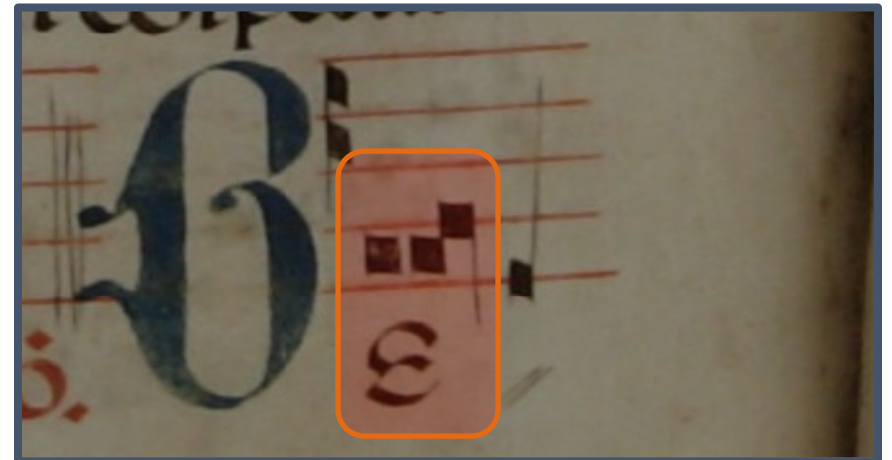
```
17 ▼ <scoreDef>
18 ▼   <staffGrp>
19     <staffDef n="1" lines="5" notationtype="neume" clef.shape="C" clef.line="4"/>
20   </staffGrp>
21 </scoreDef>
```


Add the children of layer: syllable and their children



- Add a `<syllable>` element inside the `<layer>`
- Add the following two children of `<syllable>`:
 - Add one `<syl>` element
 - Within the **syl** opening tag (`<syl>`) and its closing tag (`</syl>`), add the text of the corresponding syllable `<syl>Be</syl>`
 - Add `<neume>` elements for each neume related to the syllable (one `<neume>` in this case)

25 ▼	<code><syllable></code>
26 ▼	<code> <syl>Be</syl></code>
27	<code> <neume></neume></code>
28	<code></syllable></code>

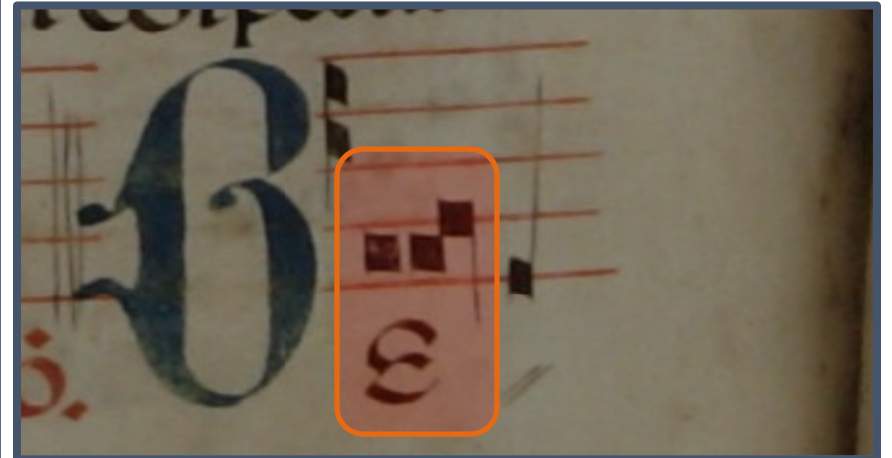


Add the children of layer: syllable and their children



- Fill in the one `<neume>` with its three neume components `<nc>`
 - For the **first** `<nc>`, add attributes `@pname = e` and `@oct = 2`
 - For the **second** `<nc>`, add attributes `@pname = e` and `@oct = 2`
 - For the **third** `<nc>`, add attributes `@pname = f` and `@oct = 2`
 - And add `@tilt = s` to the **third** `<nc>`

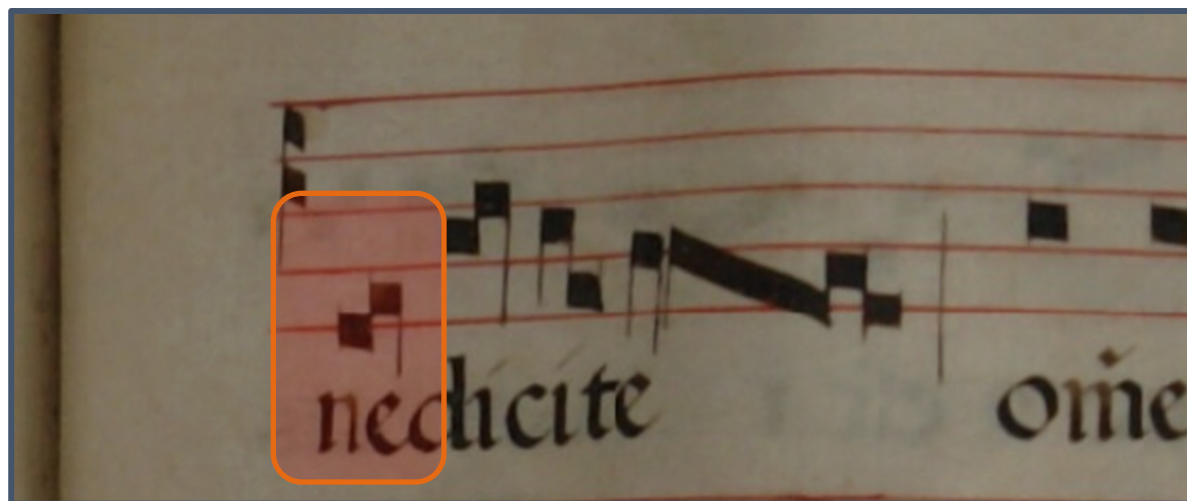
```
25 ▼ <syllable>
26 ▼   <syl>Be</syl>
27 ▼   <neume>
28     <nc pname="e" oct="2"/>
29     <nc pname="e" oct="2"/>
30     <nc tilt="s" pname="f" oct="2"/>
31   </neume>
32 </syllable>
```



Add the children of layer: syllable and their children



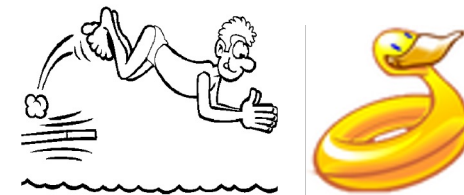
- Add another `<syllable>` element inside the `<layer>`
- Add the following two children of `<syllable>`:
 - Add one `<syl>` element
 - Within the **syl** opening tag (`<syl>`) and its closing tag (`</syl>`), add the text of the corresponding syllable `<syl>ne</syl>`
 - Add `<neume>` elements for each neume related to the syllable (one `<neume>` in this case)



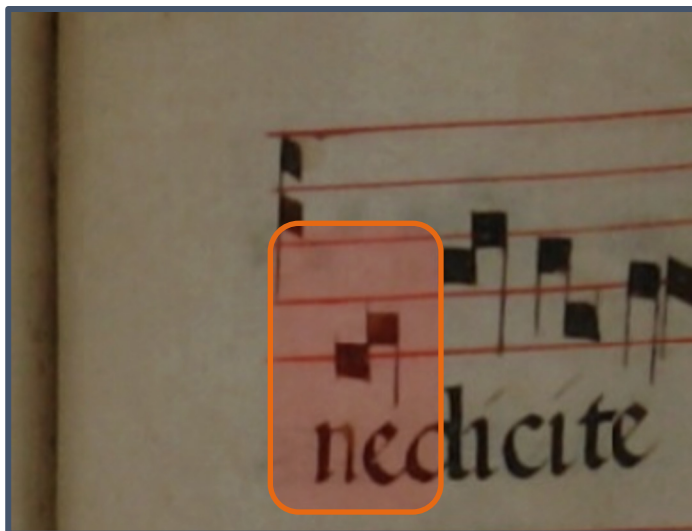
33 ▼	<code><syllable></code>
34 ▼	<code><syl>ne</syl></code>
35	<code><neume></neume></code>
36	<code></syllable></code>

nedicite omnes angeli dñi dominum

Add the children of layer: syllable and their children



- Fill in the one `<neume>` with its two neume components `<nc>`
 - For the **first** `<nc>`, add attributes `@pname = d` and `@oct = 2`
 - For the **second** `<nc>`, add attributes `@pname = e` and `@oct = 2`
 - And add `@tilt = s` to the **second** `<nc>`

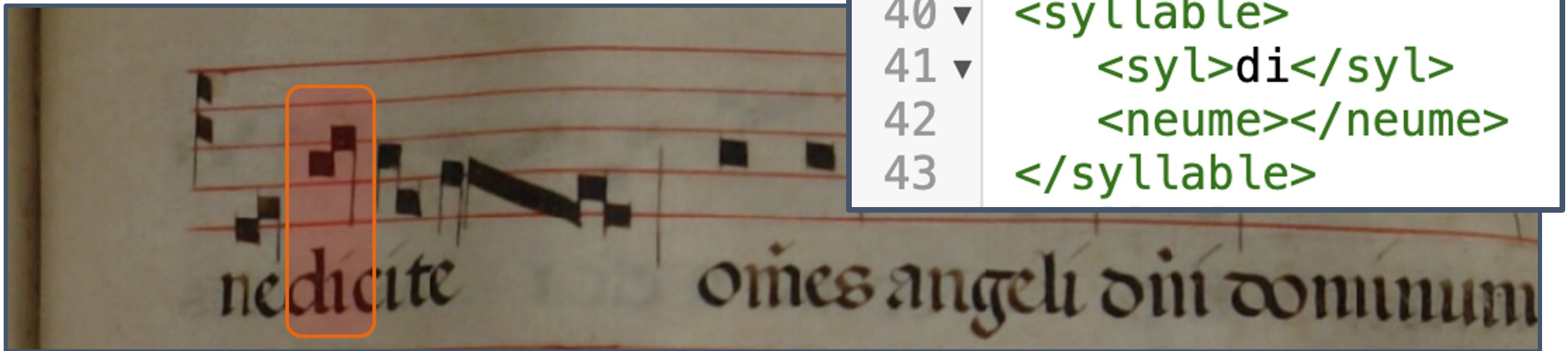


```
33 ▼ <syllable>
34 ▼   <syl>ne</syl>
35 ▼   <neume>
36     <nc pname="d" oct="2"/>
37     <nc tilt="s" pname="e" oct="2"/>
38   </neume>
39 </syllable>
40 ▼ <syllable>
```


Add the children of layer: syllable and their children



- Add another `<syllable>` element inside the `<layer>`
- Add the following two children of `<syllable>`:
 - Add one `<syl>` element
 - Within the **syl** opening tag (`<syl>`) and its closing tag (`</syl>`), add the text of the corresponding syllable `<syl>di</syl>`
 - Add `<neume>` elements for each neume related to the syllable (one `<neume>` in this case)

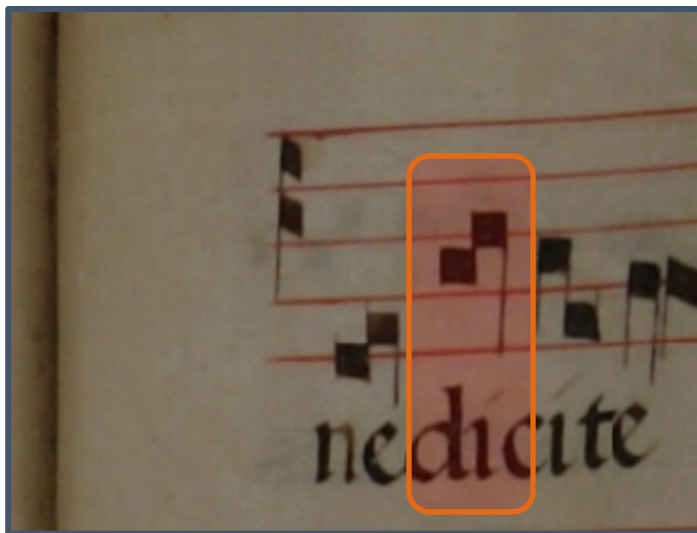


40 ▾	<code><syllable></code>
41 ▾	<code> <syl>di</syl></code>
42	<code> <neume></neume></code>
43	<code></syllable></code>

Add the children of layer: syllable and their children



- Fill in the one `<neume>` with its two neume components `<nc>`
 - For the **first** `<nc>`, add attributes `@pname = g` and `@oct = 2`
 - For the **second** `<nc>`, add attributes `@pname = a` and `@oct = 2`
 - And add `@tilt = s` to the **second** `<nc>`

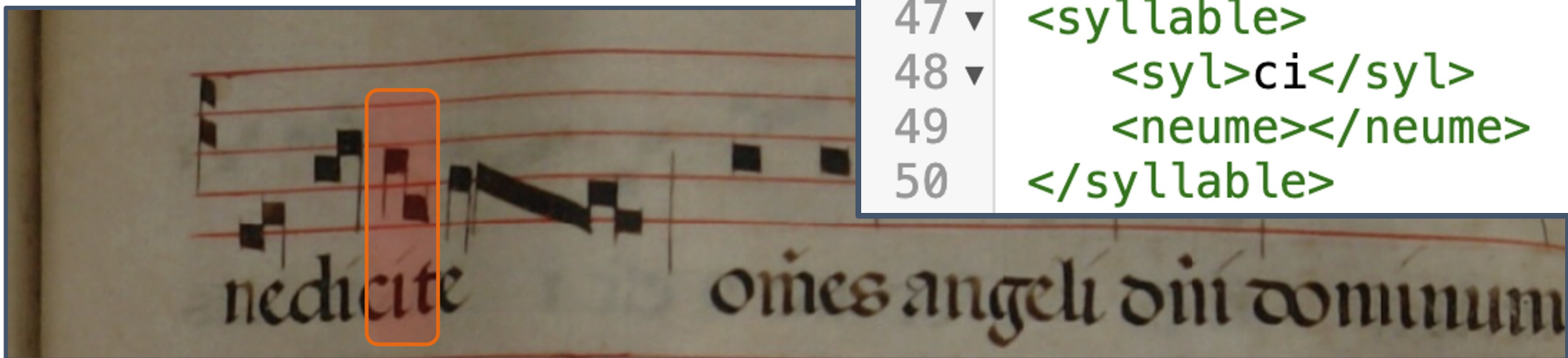


```
40 ▾ <syllable>
41 ▾   <syl>di</syl>
42 ▾   <neume>
43     <nc pname="g" oct="2"/>
44     <nc tilt="s" pname="a" oct="2"/>
45   </neume>
46 </syllable>
```

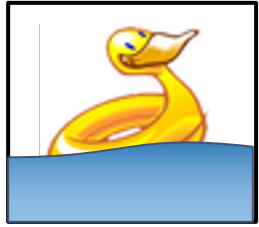

Add the children of layer: syllable and their children



- Add another `<syllable>` element inside the `<layer>`
- Add the following two children of `<syllable>`:
 - Add one `<syl>` element
 - Within the **syl** opening tag (`<syl>`) and its closing tag (`</syl>`), add the text of the corresponding syllable `<syl>ci</syl>`
 - Add `<neume>` elements for each neume related to the syllable (one `<neume>` in this case)

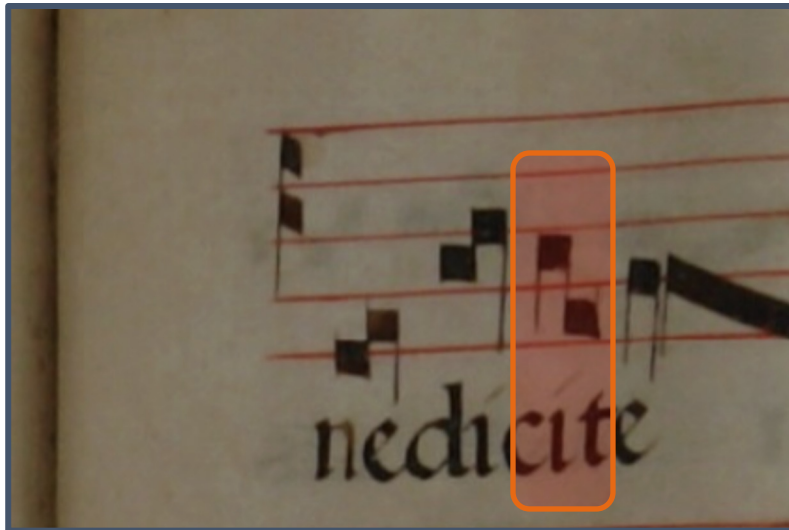


47 ▼	<code><syllable></code>
48 ▼	<code><syl>ci</syl></code>
49	<code><neume></neume></code>
50	<code></syllable></code>



Add the children of layer: syllable and their children

- Fill in the one `<neume>` with its two neume components `<nc>`
 - For the **first** `<nc>`, add attributes `@pname = g` and `@oct = 2`
 - For the **second** `<nc>`, add attributes `@pname = e` and `@oct = 2`
 - And add `@tilt = n` to the **first** `<nc>`



```
47 ▼ <syllable>
48 ▼   <syl>ci</syl>
49 ▼   <neume>
50       <nc tilt="n" pname="g" oct="2"/>
51       <nc pname="e" oct="2"/>
52   </neume>
53 </syllable>
```

omnes angeli dñi domini

Add the children of layer: syllable and their children



- Add another `<syllable>` element inside the `<layer>`
- Add the following two children of `<syllable>`:
 - Add one `<syl>` element
 - Within the `syl` opening tag (`<syl>`) and its closing tag (`</syl>`), add the text of the corresponding syllable `<syl>te</syl>`
 - Add `<neume>` elements for each neume related to the syllable (**two** `<neume>` elements in this case)

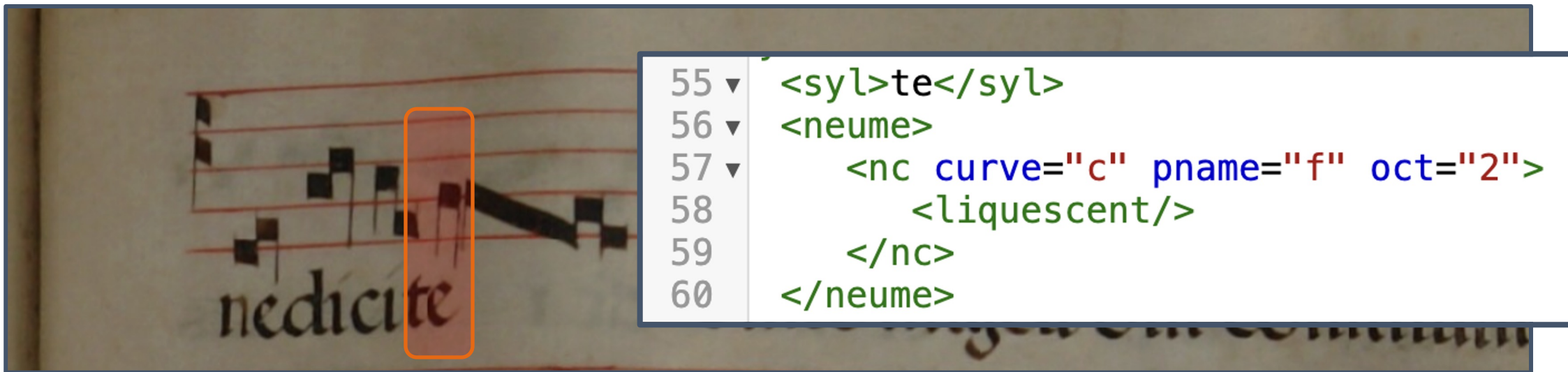
A photograph of a manuscript page with musical notation on red staves. The word 'medicite' is written in a Gothic script. A red rectangular box highlights a specific neume (a group of notes) on the staff.

54 ▼	<code><syllable></code>
55 ▼	<code><syl>te</syl></code>
56	<code><neume></neume></code>
57	<code><neume></neume></code>
58	<code></syllable></code>

Add the children of layer: syllable and their children



- Fill in the **first** `<neume>` with one neume component `<nc>` (the liquescent shape shown in the manuscript)
 - For the `<nc>`, add the pitch-related attributes `@pname = f` and `@oct = 2`
 - Then add `@curve = c`
 - Finally, add a child `<liquescent>` to the `<nc>`

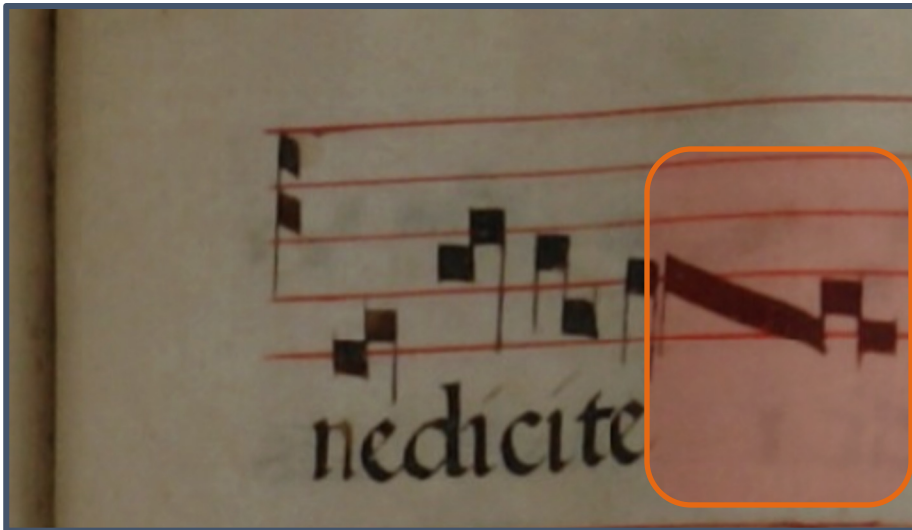


```
55 ▼ <syl>te</syl>
56 ▼ <neume>
57 ▼   <nc curve="c" pname="f" oct="2">
58       <liquescent/>
59   </nc>
60 </neume>
```




Add the children of layer: syllable and their children

- Fill in the **second** `<neume>` with its four neume components `<nc>`
 - For the **first** `<nc>`, add attributes `@pname = f` and `@oct = 2`
 - For the **second** `<nc>`, add attributes `@pname = d` and `@oct = 2`
 - For the **third** `<nc>`, add attributes `@pname = e` and `@oct = 2`
 - For the **fourth** `<nc>`, add attributes `@pname = d` and `@oct = 2`
 - Add `@tilt = n` to the **first** `<nc>`
 - And add `@ligated = true` to the **first** and **second** `<nc>`



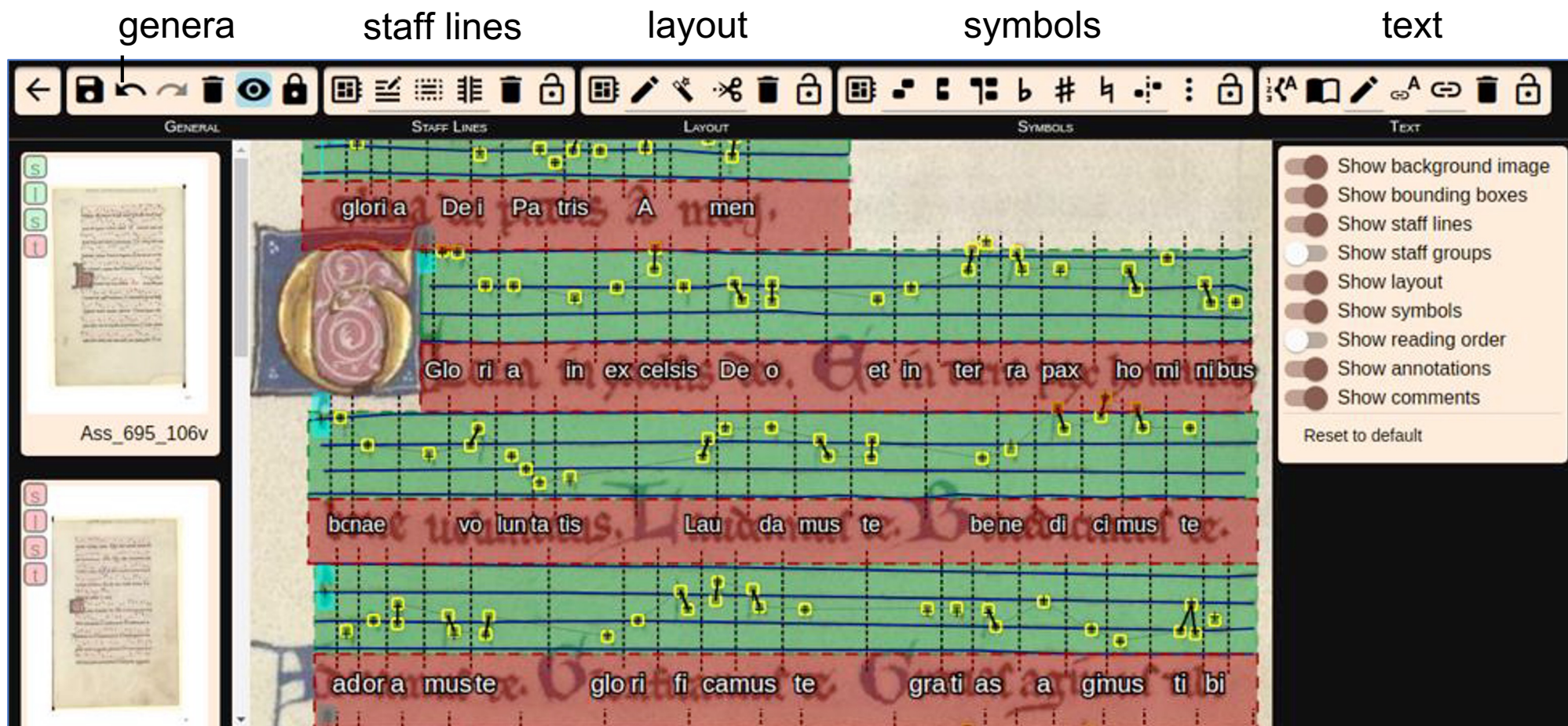
```
61 ▾ <neume>
62   <nc ligated="true" tilt="n" pname="f" oct="2"/>
63   <nc ligated="true" pname="d" oct="2"/>
64   <nc pname="e" oct="2"/>
65   <nc pname="d" oct="2"/>
66 </neume>
```


Some OMR Frameworks for Neume Scripts

OMMR4all

SIMSSA | Single Interface for Music
| Score Searching and Analysis

OMM4All



OMMR4All was created by **Cristopher Wick** (University of Würzburg), and it is being used in the **Corpus Monodicum** project

Alexander Hartelt & Jan Hajič: Collaborative digital editions with OMMR4All (DHEMR 2021/1 sess. 3): <https://youtu.be/hX9pG0dfbZ8?feature=shared>

OMR Workflow used at the **Single Interface for Music Score Searching & Analysis (SIMSSA)** Project

The screenshot displays the SIMSSA project interface, which is a web-based application for music score analysis. It is divided into three main sections:

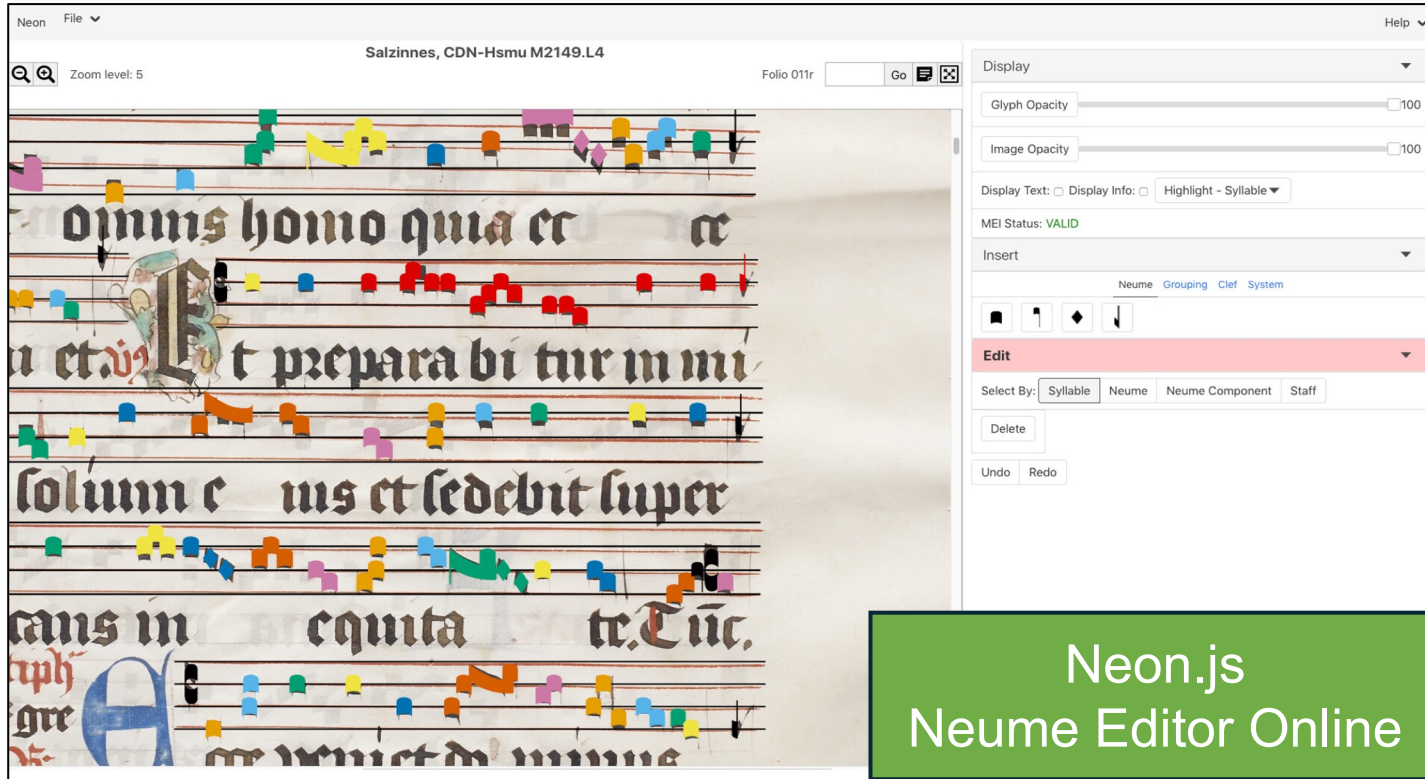
- Pixel.js:** The top-left section shows a scanned image of a manuscript page with musical notation and Latin text. Below the image is a toolbar with various editing tools like 'undo', 'redo', 'delete selected layer', and 'create new layer'. A purple box labeled 'Pixel.js' is overlaid on the bottom right of this section.
- Interactive Classifier:** The top-right section shows a grid of musical notes and symbols, each with a small colored box (green, yellow, or red) indicating its classification. A brown box labeled 'Interactive Classifier' is overlaid on the bottom left of this section.
- Neon.js:** The bottom-right section shows a zoomed-in view of the manuscript page with musical notation and Latin text. A green box labeled 'Neon.js' is overlaid on the bottom right of this section.

The bottom-left corner of the interface features the **RODAN** logo, which consists of an orange circle with a white bird-like shape inside, followed by the word 'RODAN' in a sans-serif font.

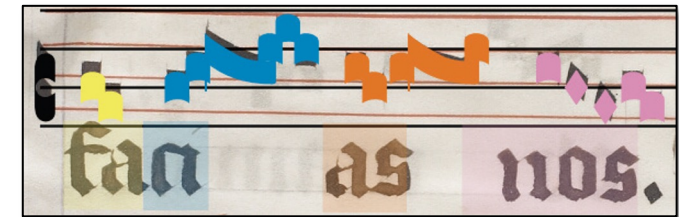
Martha Thomae: Making Rodan work for you (DHEMR 2021/1, Session 2)
https://youtu.be/_TeGXG9Fh2M?si=v_IQOByslGi7GtLL

The **SIMSSA** project is
directed by **Ichiro Fujinaga**
(McGill University)

OMR Workflow used at the **Single Interface for Music Score Searching & Analysis (SIMSSA)** Project



Neon.js
Neume Editor Online



Martha Thomae: Making Rodan work for you (DHEMR 2021/1, Session 2)
https://youtu.be/_TeGXG9Fh2M?si=v_IQOByslGi7GtLL

The **SIMSSA** project is
directed by **Ichiro Fujinaga**
(McGill University)

Resources

<https://music-encoding.org/>

- Tutorials on MEI: <https://music-encoding.org/resources/tutorials.html>
- Guidelines → Chapter 6: Neume Notation (<https://music-encoding.org/guidelines/v5/content/index.html>)
- Elements reference: <https://music-encoding.org/guidelines/v5/elements.html>

MEI Tutorials

On this page, you'll find a number of small tutorials for MEI, each

If you're about to start learning MEI, we recommend to start with the 5 minute Quickstart tutorial which will let you encode a very simple melody with MEI.

If you're about to write a new MEI tutorial, we recommend to start with the Advanced tutorial which will provide backgrounds about the necessary steps. Please also have a look at our list of missing tutorials on Github.

External tutorials & related material
For community-created resources including tutorials and teaching materials, see [Pedagogy & Praxis Resources](#).

For a Spanish translation of the tutorials, see [Tutoriales en español](#).

<https://music-encoding.org/resources/tutorials.html>

Minimal structure of a valid MEI file

Encoding your first melody

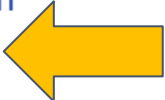
Structure of all the encoding within <music>

Available Tutorials

- BEGINNERS: XML basics and minimal MEI structure**
A short tutorial about the basics of XML & MEI
- BEGINNERS: Quickstart**
A 5-minute Quickstart tutorial for MEI
- ADVANCED: Incipit encoding**
An advanced tutorial on how to encode incipits in MEI
- BEGINNERS: Chords**
A short tutorial about chords in MEI
- BEGINNERS: Rests**
A short tutorial about rests in MEI
- ADVANCED: Understanding ODD**
An advanced tutorial on the internal structure of MEI
- ADVANCED: Customizing MEI (WIP)**
Creating custom MEI profiles
- META: Writing Tutorials**
An Introduction to writing MEI tutorials

MEI Guidelines (5.0)

- 1 Introduction to MEI
- 2 Shared Concepts in MEI
- 3 Metadata in MEI
- 4 Repertoire: Common Music Notation
- 5 Repertoire: Mensural Notation
- 6 Repertoire: Neume Notation
- 7 Repertoire: String Tablature
- 8 Lyrics and Performance Directions
- 9 Text Encoding
- 10 Analysis Markup and Harmonies
- 11 Scholarly Editing with MEI
- 12 Facsimiles and Recordings
- 13 Linking Data
- 14 Integrating MEI with other Standards and Formats



MEI Version: 5.0 (#eb14650)

🔍

Guidelines

Modules

Elements

Model Classes

Macro Groups

Attribute Classes

Data Types



6 Repertoire: Neume Notation

This chapter describes the elements, model classes, and attribute classes that are part of the MEI.neumes module.

6.1 Overview of the Neumes Module

The MEI Neumes Module represents the community's attempt to create a standardized set of rules that encapsulate in a logical, systematic, and unequivocal way the musical information represented and conveyed by Western European neumatic notations (beginning with the late ninth century and continuing to the printed books of the twentieth). Most neume notation is used to set music to an existing text. The syllable is the fundamental unit of structure, with the neumes themselves serving as a means of "sonifying" the text. A syllable may be expressed via one or more neumes, with the particular neume shape chosen depending on the pitch contour that is being employed and the desired interpretation.

The `syllable` element is used as the primary organizational element for neume notation within a `layer` element. Within `syllable`, the `syl` element defined in the `MEI.shared` module is used for encoding the textual content, while the `neume` and `nc` elements are used to encode the neumes themselves. Within these Neumes Module elements, other standard MEI mechanisms are available to accommodate, for example, editorial or critical markup.

6.1.1 Basic four elements

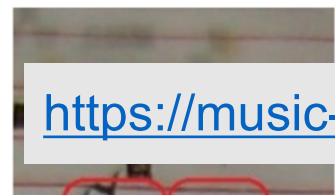
The following four elements are the fundamental components of the Neumes Module:

`<syllable>`
Neume notation can be thought of as "neumed text". Therefore, the syllable element provides high-level organization in this repertoire.

Neume notation can be thought of as "neumed text". Therefore, the syllable element provides high-level organization in this repertoire.

`<syl>`
Individual lyric syllable.

(syllable) – Individual lyric syllable.



<https://music-encoding.org/guidelines/v5/content/neumes.html>

MEI Version: 5.0 (#eb14650)

- Guidelines
- Modules
- Elements
- Model Classes
- Macro Groups
- Attribute Classes
- Data Types

6.1 Overview of the Neumes Module

6.1.1 Basic four elements

6.2 Neumes Module Background

6.3 Neume Notation and MEI

6.4 Samples of MEI encodings

- 6.4.1 Elements
- 6.4.2 Neume component attributes
- 6.4.3 Custos attributes
- 6.4.4 Episema attributes
- 6.4.5 Liquescent attributes
- 6.4.6 Old Hispanic tick attributes
- 6.4.7 Quilisma attribute
- 6.4.8 Significant letters attribute
- 6.4.9 Note
- 6.4.10 Basic Encoding – Syllable
- 6.4.11 Manuscripts

6.5 Bibliographic References

music-encoding.org/guidelines/v5/elements/nc.html#attributes_compact_tab

MEI

AboutCommunityConferenceResourcesDocumentationArchive

<nc>

Sign representing a single pitched event, although the exact pitch may not be known.

Referenced in Chapters:

6.1.1 Basic four elements, 6.4.1 Elements, 6.4.2 Neume component attributes, 6.5 Bibliographic References, 10.1.2.6 Solmization

Module

MEI.neumes

Attributes

compact

full definitionby classby module

@accid.ges, @altsym, @angled, @artic.ges, @class, @color, @con, @copyof, @corresp, @curve, @deg, @dots.ges, @dur.ges, @dur.metrical, @dur.ppq, @dur.real, @dur.recip, @fac, @follows, @fontfam, @fontname, @fontsize, @fontstyle, @fontweight, @glyph.auth, @glyph.name, @glyph.num, @glyph.uri, @ho, @hooked, @instr, @intm, @label, @layer, @letterspacing, @ligated, @lineheight, @loc, @mfunc, @n, @next, @oct, @oct.ges, @pclass, @pname, @pname.ges, @pnum, @precedes, @prev, @psolfa, @rellen, @resp, @s-shape, @sameas, @staff, @synch, @tilt, @type, @vel, @visible, @when, @x, @xml:base, @xml:id, @y

Member of

model.neumePart

Groups elements that may occur within a neume.

Contained By

compact

by classby module

<abbr>, <corr>, <damage>, , <expan>, <lem>, <ncGrp>, **<neume>**, <orig>, <rdg>, <reg>, <restore>, <sic>, <supplied>, <unclear>

May Contain

compact

by classby module

<add>, <app>, <choice>, <corr>, <damage>, , <episema>, <gap>, <handShift>, <hispanTick>, **<liquescent>**, <orig>, **<oriscus>**, **<quilisma>**, <reg>, <restore>, <sic>, <signifLet>, <strophicus>, <subst>, <supplied>, <unclear>

MEI Version: 5.0 (#eb14650)

Guidelines

Modules

Elements

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Attribute Classes

Data Types

https://music-encoding.org/guidelines/v5/elements/nc.html#attributes_compact_tab

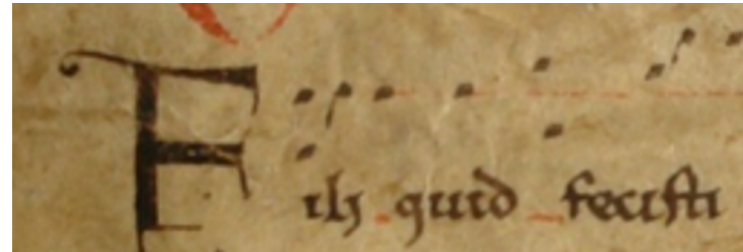
72



```

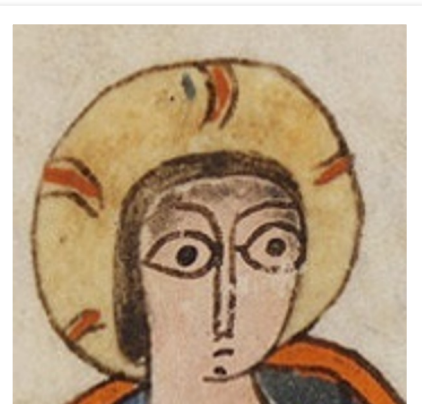
<syllable xml:id="m-43558eee-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-43558f20-8a6a-11ee-a3f2-3645f29ecd15">Fi</syl>
  <neume xml:id="m-43558f52-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-43558f8e-8a6a-11ee-a3f2-3645f29ecd15"/>
    <nc loc="-2" xml:id="m-43558fc0-8a6a-11ee-a3f2-3645f29ecd15"/>
    <nc loc="0" tilt="ne" xml:id="m-43558ff2-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>
<syllable xml:id="m-43559024-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-4355904c-8a6a-11ee-a3f2-3645f29ecd15">li</syl>
  <neume xml:id="m-4355907e-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-435590b0-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>
<syllable xml:id="m-435590d8-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-4355910a-8a6a-11ee-a3f2-3645f29ecd15">quid</syl>
  <neume xml:id="m-4355913c-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-43559178-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>

```



Thanks also to
Laurent Pugin (Verovio),
David Weigl and Werner
Goebl (mei-friend),
 and **Anna Plaksin (mei-**
 friend, some slides, and the
 “ducky”)

Thank you!





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UNIVERSIDADE NOVA DE LISBOA

Thank you!

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elsadeluca@fcsh.unl.pt



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